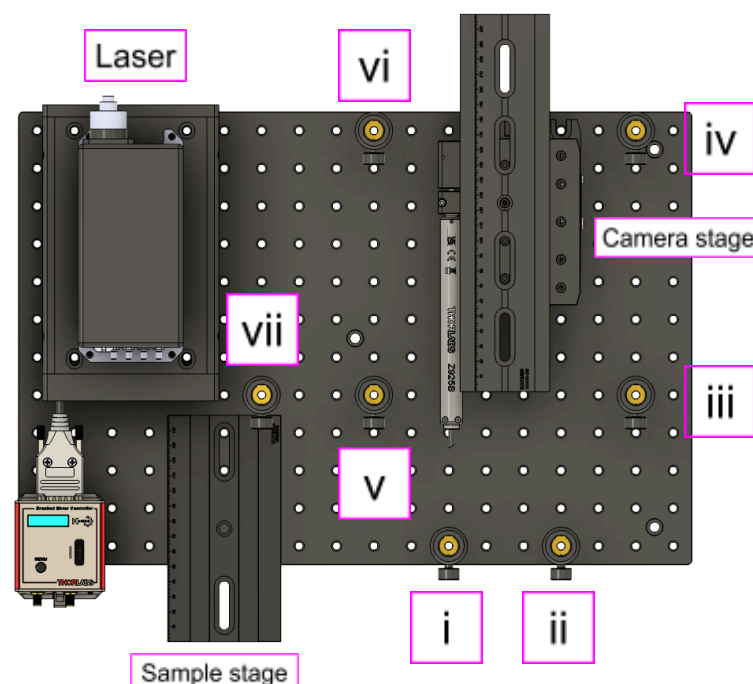


descSPIM-Fullmoon: Construction procedures

1. Construction

- 1-1. Stage setup and positioning
 - 1-1-1. Installation of motorized actuator
 - 1-1-2. Component Positioning on the Breadboard
- 1-2. Construction of the excitation optics
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 - 1-2-4. Light source assembly
- 1-3. Constructing of the detection optics
- 1-4. Construction of the sample holder

2. Optical alignment of descSPIM-Fullmoon

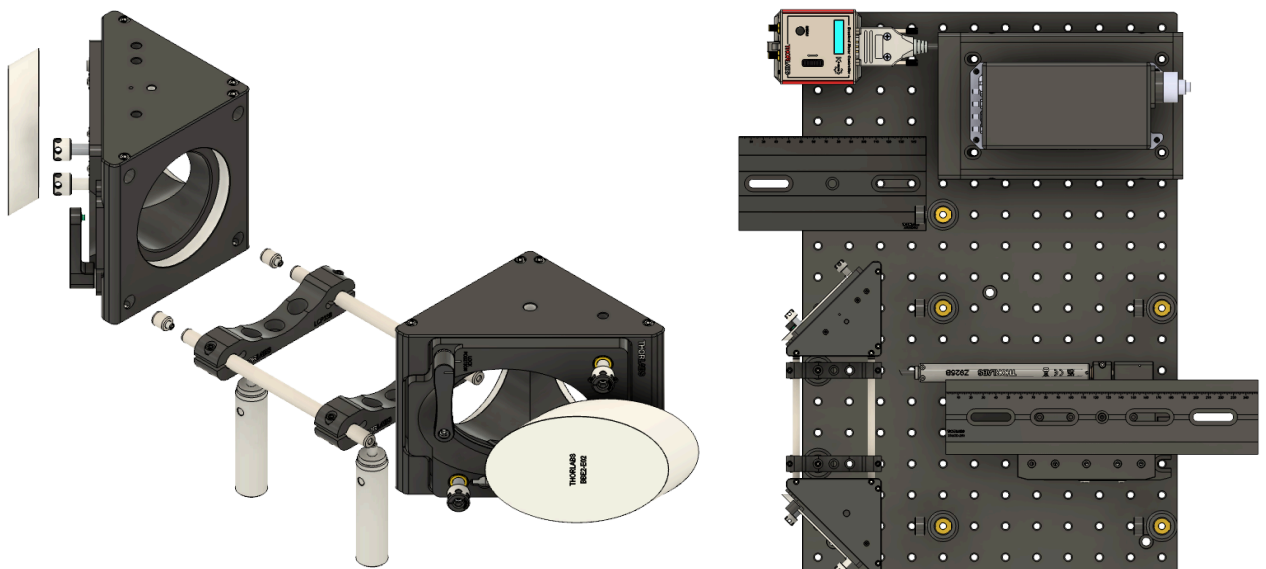


1-2. Construction of the excitation optics

1-2-1. Integrating a 2-Inch Mirror

Mount E02 Broadband Dielectric-Coated Elliptical Mirrors (BBE2-E02 x2, Thorlabs) onto 60 mm Cage Right-Angle Kinematic Elliptical Mirror Mounts (KCB2EC/M x2, Thorlabs). Secure Ø12.7 mm posts (TR50/M x2, Thorlabs) to the side holes of the two 60 mm Cage Mounting Brackets (LCP33B x2, Thorlabs) using the included M4 screws. [CAUTION: Use Ø12.7 mm posts instead of Ø12 mm. Using Ø12 mm posts can result in uncontrolled optical distortion due to a 0.7 mm gap.]

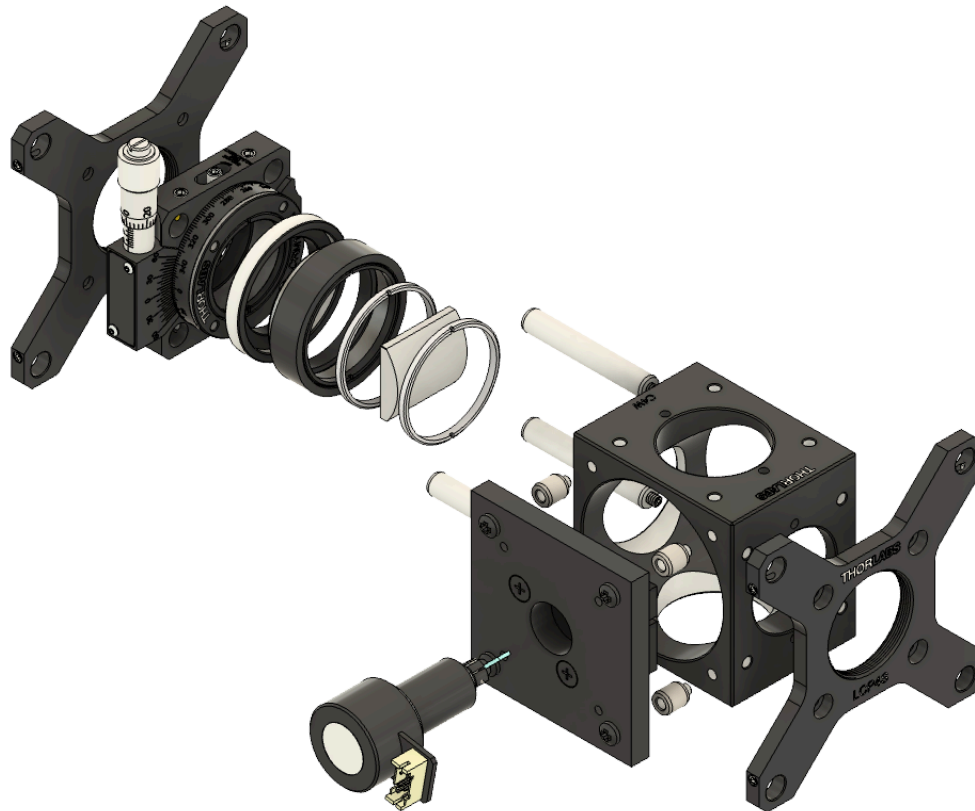
Insert two Ø6 mm Cage Assembly Rods (Thorlabs, ER4 + ER025 x2) through the brackets and attach the mirror mounts to both ends of the rods. [CAUTION: Pay close attention to the orientation of the mirror holders. If the mirrors are not arranged in a left–right symmetric configuration, the two illumination paths will not be properly aligned.] Insert these two posts into the holders (i and ii) on the breadboard.



1-2-2. Right Optical Path Assembly

Remove the plate for inserting the galvanometric mirror from a 30 mm Cage System Adapter (GCM001, Thorlabs). Screw four rods (ER025 x4, Thorlabs) into the threaded holes on the box side, and secure the removed plate to the rods using screws (SH4S031, Thorlabs). Attach a 30 mm to 60 mm Cage Plate Adapter (LCP4S, Thorlabs) to the box using the included screws, and screw three rods (ER1.5 x3, Thorlabs) into the opposite side. Slide a Precision 30 mm Cage Rotation Mount (CRM1PT/M, Thorlabs) onto the rods, then secure a 30 mm to 60 mm Cage Plate Adapter (LCP4S, Thorlabs) to the rods using the included screws. Fix the components in the SM30 Lens Tube (SM30L03, Thorlabs) in the following order:

Standard Retaining Ring (included), Plano-Convex Cylindrical Lens (LJ1212L1-A, Thorlabs), and Standard Retaining Ring (SM30RR, Thorlabs). Then attach the lens tube to the Rotation Mount via an Adapter (SM1A16, Thorlabs).



Insert rods (ER10 x3, Thorlabs) through the 60 mm cage holes of the two 30 mm to 60 mm Cage Plate Adapters, and slide a Cylindrical Lens Mount (CYLCP/M, Thorlabs) with Plano-Convex Cylindrical Lens (LJ1640L1-A, Thorlabs). Pass the rod (ER4 + ER0.5, Thorlabs) through the corner where the knob of the rotation mount is located. [CAUTION: Ensure that the rod of the cylindrical lens holder is positioned on the same side as the galvanometric mirror.]

Screw four rods (ER3 x4, Thorlabs) into the top side of the galvanometric adapter box. Set mirrors (BBE1-E02, Thorlabs) into 30 mm Cage Right-Angle Kinematic Elliptical Mirror Mounts (KCB1EC/M, Thorlabs), insert the rods into the through-holes, and secure them. Four rods (ER3 + ER05, Thorlabs) were inserted into the through-holes of the mirror mounts and fixed with screws. A Cage Rotation Mount (CRM1T, Thorlabs) was mounted onto the rods. A mirror mount (KCB1EC/M, Thorlabs) with a mirror (BBE1-E02, Thorlabs) was then attached to the rods and fixed in place. A post holder (PH30/M, Thorlabs) was attached to the bottom of the mirror mount using an M6 set screw, and a post (TR40/M, Thorlabs) was inserted

and fixed. A Snap-On 60 mm Cage Mounting Bracket (LCPMA4/M, Thorlabs) was attached to the upper rod of the components assembled. The post holder attached to the bottom of the mirror mount was loosened, and the post was rotated and mounted onto the bracket.



Inject a collimated alignment laser beam into the mirror mount (KCB1EC/M), and use a 30 mm Cage Alignment Plate (CPA1, Thorlabs) to adjust the two mirror mounts (KCB1EC/M) so that the laser beam passes through the center of the optical path at the top of the galvanometric mirror box. Adjust the position of the galvanometric mirror so that it is centered on the laser beam. Set the galvanometric mirror to the 0 V position, place a 60 mm Cage Alignment Plate (LCPA1, Thorlabs) at the focal point of the Plano-Convex Cylindrical Lens (LJ1212L1-A, Thorlabs), and adjust the mirror angle so that the excitation beam passes through the center, then fix the mirror in place. Place a Shearing Interferometer (SI254 + SICP + SM1T4 + LCP33/M,

Thorlabs) after the Cylindrical Lens Mount (CYLCP/M, Thorlabs) and adjust the height of the Plano-Convex Cylindrical Lens (LJ1640L1-A, Thorlabs) and the position of the Cylindrical Lens Mount so that the alignment beam becomes collimated. Fix the position of the Cylindrical Lens Mount using the rod (ER4 + ER0.5, Thorlabs). A $f = 500$ cylindrical lens (LJ1144RM-A, Thorlabs) was placed in a lens tube (SM1L03, Thorlabs) and secured in a Cage Rotation Mount (CRM1T, Thorlabs) that was installed. Attach a laser blade as a slit to the top surface of the galvanometric mirror box to prevent the excitation beam from hitting the edge of the galvanometric mirror. [CAUTION: Perform this alignment step at this stage, as carrying it out later becomes extremely cumbersome.]

Mount two $\varnothing 12.7$ mm posts (TR50/M, Thorlabs) equipped with Snap-On 60 mm Cage Mounting Brackets (LCPMA4/M, Thorlabs) onto the lower rod of this component. Insert these two posts into the holders on the breadboard (iii and iv). [CAUTION: Fix the posts while ensuring that the bracket and the holder are firmly in contact.]

Connect the three rods (ER10 x3, Thorlabs) to the mirror holder installed on the breadboard and secure them. [CAUTION: Do not change the position of the short rod (ER4 + ER0.5), as they determine the position of the cylindrical lens mount.]

1-2-3. Left Optical Path Assembly

Repeat the steps for the right optical path to assemble the second galvo components, ensuring that the galvanometric mirror is oriented opposite to the previously assembled one.

Insert rods (ER10 x3, Thorlabs) through the 60 mm cage holes of the two 30 mm to 60 mm Cage Plate Adapters, and slide a 60 mm Cage Mounting Bracket (LCP33C/M, Thorlabs) and a Cylindrical Lens Mount (CYLCP/M, Thorlabs) with Plano-Convex Cylindrical Lens (LJ1640L1-A, Thorlabs). [CAUTION: Ensure that the rod of the cylindrical lens holder is positioned on the same side as the galvanometric mirror.] Pass the rod (ER4 + ER0.5, Thorlabs) through the corner where the knob of the rotation mount is located. [CAUTION: Do not forget to insert the 60 mm Cage Mounting Bracket onto the upper rod; inserting it later is extremely difficult.]

Screw four rods (ER3 x4, Thorlabs) into the top side of the galvanometric adapter box. Set mirrors (BBE1-E02, Thorlabs) into 30 mm Cage Right-Angle Kinematic Elliptical Mirror Mounts (KCB1EC/M, Thorlabs), insert the rods into the through-holes, and secure them. Four rods (ER3 + ER05, Thorlabs) were inserted into the through-holes of the mirror mounts and fixed with screws. A Cage Rotation Mount (CRM1T, Thorlabs) was mounted onto the rods.

A beam splitter (CCM1-BS013/M, Thorlabs) was then attached to the rods and fixed in place. A post holder (PH30/M, Thorlabs) was attached to the bottom of the beam

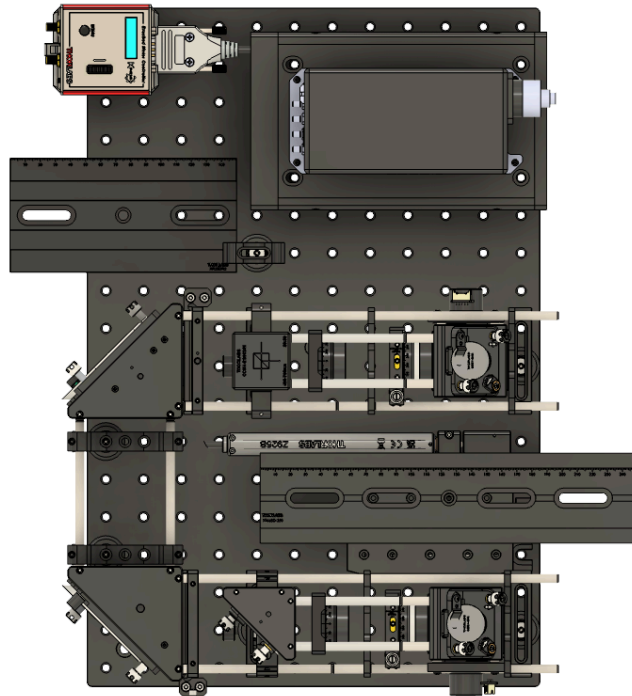
splitter using a Thread Adapter (AP6M4M, Thorlabs), and a post (TR40/M) was inserted and fixed. The post holders attached to the beam splitter were loosened, and the posts were rotated and mounted onto the bracket installed.



Perform optical alignment in the same manner as for the right side. [CAUTION: Perform this alignment step at this stage, as carrying it out later becomes extremely cumbersome.]

Mount two $\varnothing 12.7$ mm posts (TR50/M, Thorlabs) equipped with Snap-On 60 mm Cage Mounting Brackets (LCPMA4/M, Thorlabs) onto the lower rod of this component. Insert these two posts into the holders on the breadboard (v and vi). [CAUTION: Fix the posts while ensuring that the bracket and the holder are firmly in contact.]

Connect the three rods (ER10 x3, Thorlabs) to the mirror holder installed on the beadboard and secure them. [CAUTION: Do not change the position of the short rod (ER4 + ER0.5), as they determine the position of the cylindrical lens mount.]



1-2-4. Light source assembly

Insert Ø12.7 mm posts (TR100/M, Thorlabs) with a Snap-On 30 mm Cage Mounting Bracket (CPMA4/M, Thorlabs) into the 150 mm holder on the breadboard (vii).

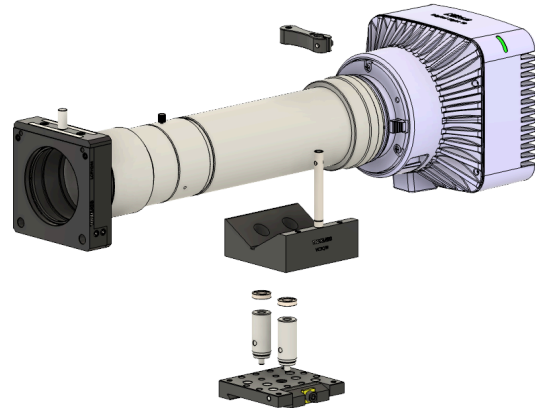
Four rods (ER6 + ER1.5, Thorlabs) were fixed to the beam splitter, and a XY Translating Lens Mount (CXY1, Thorlabs) with a $f = 100$ achromatic collimating lens (AC254-100-A, Thorlabs) and a cage plate (CP33/M, Thorlabs) with a fiber adapter (SM1FC, Thorlabs) were passed through the rods.

The post (TR100/M, Thorlabs) was lifted, and the bracket (CPMA4/M, Thorlabs) was pressed onto the rod until it clicked into place (Loosen the screws on both sides of the brackets as well as the screws securing the posts). Tighten all screws.



1-3. Constructing of the detection optics

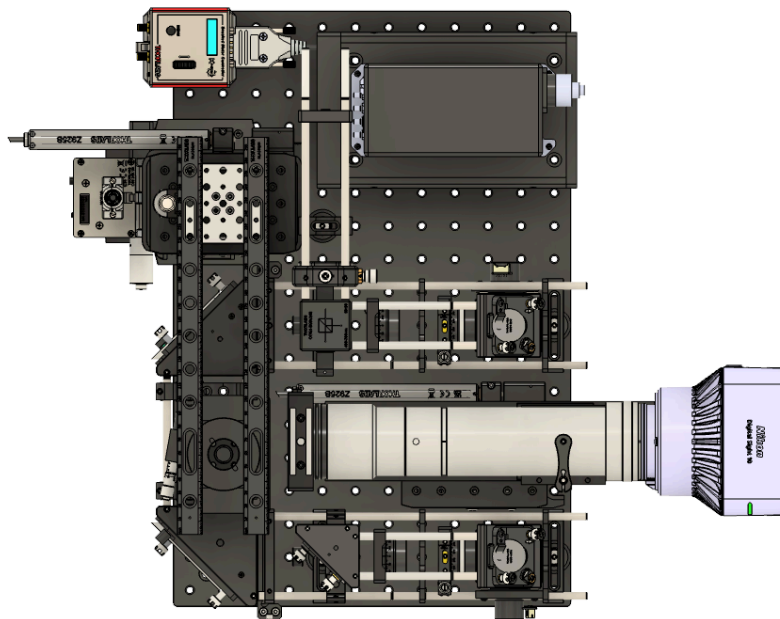
The 60 mm Filter Holder (LCFH2/M, Thorlabs) was connected to the large-format $\phi 48$ mm telecentric lens (LSTL10H-F, Myu-tron) via the adapters (SM2T2, Thorlabs / SM2A53, Thorlabs / SKU0629, K&F Concept) and attached to the CMOS camera (DS50M, Nikon).



Secure the two $\text{Ø}12.7$ mm posts (TR50/M, Thorlabs) to a Rail Carriage (XT66P2/M, Thorlabs) using M4 set screws.

Secure the $\text{Ø}12.7$ mm Post Spacers (TR3M, Thorlabs) and the A Large V-Clamp (VC3C/M, Thorlabs) to the posts using M6 screws. Secure the lens to the V-Clamp.

Place the base plate onto the rail of the camera stage installed on the breadboard.

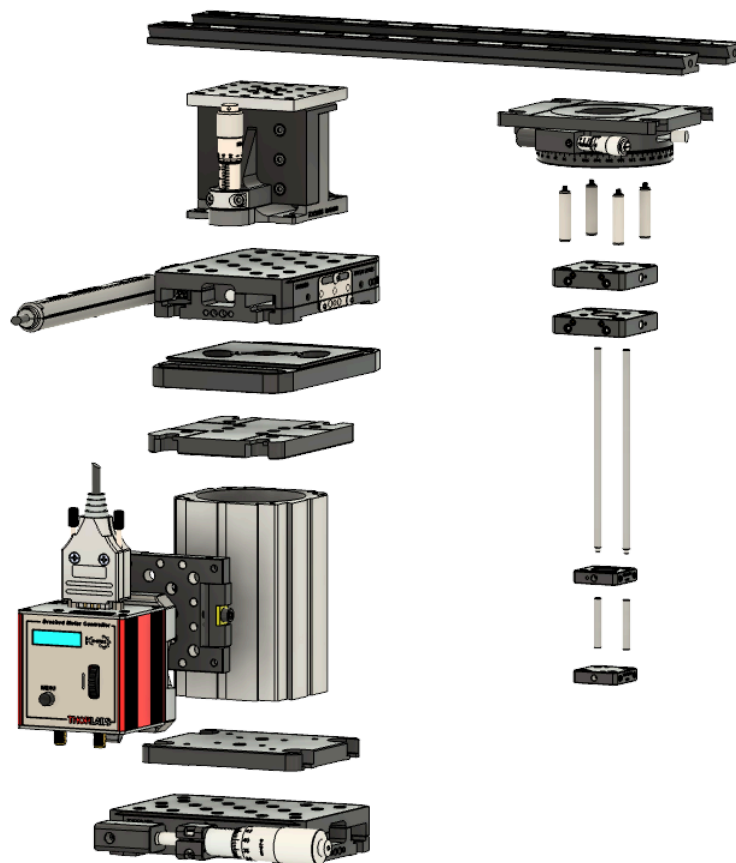


1-4. Construction of the sample holder

Screw four rods (ER1, Thorlabs) into the 30 mm cage holes of the Rotation Stage (PR01/M, Thorlabs) and secure them. Insert two Cage Adapter Plates (SP15/M, Thorlabs) onto the rods and fix them in place. Insert two rods (SR6 + SR1, Thorlabs) into these Cage Adapter Plates and secure them. Mount two Blank Plates (SP01, Thorlabs) onto the rods and fix them.

Attach the Vertical Mounting Plates (XT66P1, Thorlabs) to both ends of the 66 mm Construction Rail (XT66-100, Thorlabs) using the supplied screws. Mount the assembly onto the stage (XR25P/M, Thorlabs) using M6 screws. Secure the Base Plate (XR25-B1/M, Thorlabs) on top with M6 screws and nuts. Mount the sample stage (XR25P/M, Thorlabs) equipped with the motorized actuator (Z825B, Thorlabs) onto the Base Plate and secure it. Fix the Vertical Translation Stage (MVS05/M, Thorlabs) onto this stage using M6 screws. Attach two Dovetail Optical Rails (RLA150/M, Thorlabs) onto the Vertical Translation Stage with M6 screws.

Attach this assembly to the rail constructed. Then mount it onto the rail installed on the breadboard.



2. Optical alignment of descSPIM-Fullmoon

Use a Shearing Interferometer to adjust the distance between the collimating lens and the fiber adapter (SM1FC, Thorlabs) so that the laser beam becomes collimated. Using a 30 mm Cage Alignment Plate (CPA1, Thorlabs), adjust the XY Translating Lens Mount (CXY1, Thorlabs) and the mirror mounts (KCB1EC/M, Thorlabs) so that the excitation beam collimated by the $f = 100$ lens travels through the center of the optical path and reaches the galvanometric mirror box.

Fixed rods (ER1.5) onto the 60 mm Cage Right-Angle Kinematic Mirror Mounts (KCB2EC/M), place a 60 mm Cage Alignment Plate (LCPA1, Thorlabs). Adjust the 60 mm Cage Right-Angle Kinematic Mirror Mounts and the 30 mm Cage Right-Angle Kinematic Mirror Mounts located directly above the galvanometric mirror adapter box so that the light sheet passes through the center of the optical axis.